Methods description / MVA Hackathon 2026, Track 2 (Drug Repurposing)

Describe your approach in detail (how you went from variant/mechanism to candidate medication(s)). A model-constrained repurposing screen. Starting from the Track 1 compound-heterozygous BUB1B pair (biallelic BUBR1 loss weakens the spindle-assembly checkpoint), a three-equation mass-action ODE model of the checkpoint (calibrated on BUBR1 perturbation data, independently validated, falsification documented) sizes the required effect (>= 2.2x functional BUBR1, conservative) and ranks levers (L1 BUBR1 amount = readthrough + hypomorph stabilization; L3 MCC stability, not druggable; L4 APC/C feedback, infeasible). Five filters: (1) lever - documented mechanism on L1; (2) effect size - published class effects must reach the model floor; (3) regulatory - approved in FDA/EMA/MHRA/PMDA with pediatric dosing experience, verified against primary sources (2026-10-15); (4) oncology - aneuploid-survival agents excluded (MVA1 is cancer-prone); (5) anti-target - checkpoint-impairing drugs (MPS1/TTK, Aurora B, PLK1, Eg5) excluded. Output: amlexanox (NMD inhibition + readthrough; the PTC sits 74 nt upstream of the final exon junction, so NMD is rate-limiting) plus Ravicti/4-PBA (pediatric chaperone for the destabilizing N1002K hypomorph) as a combination, because a dG bridge shows readthrough alone supplies at most ~28% of the required increment; ataluren is a readthrough-only comparator (regulatory footprint collapsing). A five-assay patient-cell validation plan with quantitative pass criteria is proposed (target: ~5-10x reduction of the aneuploidy rate in the dish).

(required) Generative AI used: provider, plan/tier, settings. Indiana University REALLMS (university self-hosted LLM endpoint) was used; non-commerical. This resource has been approved by the institution for use with PHI.

Was candidate identification automated or primarily manual? Hybrid: rule-based screening logic from the ODE model; automated ChEMBL target screen; manual final curation, and regulatory verification.

Describe any manual literature review or expert curation in detail. Primary-source regulatory audit (FDA/EMA/MHRA labels and ClinicalTrials.gov, verified 2026-10-15: ataluren EMA non-renewal and FDA NDA withdrawal; arimoclomol FDA approval and EMA refusal); literature searches for nonsense-suppression and chaperone effect sizes in patient cells from other diseases (CF, RDEB, DMD, ABCA1).

Publicly available databases and literature only, or proprietary sources too? Public sources only + challenge-provided proband VCF.

Describe the public data sources used in detail. ChEMBL (BUBR1 modulator screen - no approved or investigational scaffold/kinetochore modulator exists); ClinicalTrials.gov (amlexanox NCT01842282/NCT01975935); FDA/EMA/MHRA labels (DailyMed/EPAR/PLGB); OMIM and PubMed primary literature; AlphaMissense; ThermoMPNN stability dG; Boltz-2 co-complex structures; Pangolin regulatory-variant screen (heart/brain, no hits).

Describe any proprietary data sources used in detail. None.

How did you characterize the variant's mechanism? Biallelic BUB1B loss of function. Allele 1 c.2210T>G p.(Leu737Ter): stop-gain 74 nt upstream of the final exon junction, transcript destroyed by NMD - complete loss on that allele. Allele 2 c.3006T>G p.(Asn1002Lys): destabilizing hypomorph - BUB3/Cdc20 interface unchanged (Boltz-2) while stability is impaired (ThermoMPNN dG + 1.51 kcal/mol; AlphaMissense 0.9229) - partial loss (~15% residual function). Pathway disrupted: BUBR1/Bub3/MCC checkpoint inhibition of APC/C. Consequence: premature anaphase, chromosome mis-segregation, mosaic aneuploidy, cancer predisposition.

Estimated time or effort spent on this analysis. Several weeks of iterative work across the hackathon; cumulative ~70 hours.

Applied systems biology for a testable drug-repurposing hypothesis in mosaic variegated aneuploidy

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Abstract

Sequencing can identify candidate rare-disease variants without specifying what cellular function is lost or which treatment might help. I investigated two unphased BUB1B variants in a suspected MVA1 case: p.Leu737Ter, predicted from its transcript position to be susceptible to nonsense-mediated decay, and p.Asn1002Lys, predicted by machine learning models to destabilize BUBR1. Because phase, transcript abundance, and protein function are unmeasured, I introduced alternative molecular-defect scenarios into an implementation of the published Henze spindle assembly checkpoint network. Sensitivity and nonlinear analysis identified restoration of available BUBR1 as the leading tested intervention. I added a small attachment-error module to the published dynamical system because checkpoint signaling alone cannot measure erroneous chromosome segregation. The extension showed exploratory agreement with published human depletion observations, but the displayed model setting was selected for and is not empirically validated. In severe-depletion simulations, restoring BUBR1 improved both checkpoint signaling and modeled unresolved-attachment risk. Phenylbutyrates and amlexanox offered experimental drug hypotheses, but neither has been tested for these specific variants. To test whether the method transfers to other causes of MVA, I also ran CEP57/MVA2 and TRIP13/MVA3 pilot perturbations. These reused the existing model components but exposed different mechanistic sensitivities and biological processes still missing from the models. The immediate clinical-research output remains a series of discriminating experiments connecting genotype, functional BUBR1 abundance and chromosome-segregation fidelity.

1. Clinical question: what is wrong, and what is still unknown?

Mosaic variegated aneuploidy describes abnormal chromosome counts distributed across different cells and tissues. BUB1B is an established MVA-associated gene; it encodes BUBR1, which participates in the spindle assembly checkpoint and in kinetochore attachment regulation. The clinical challenge is not simply to nominate a drug associated with this gene. It is to determine what the variants do to BUBR1 and which intervention could reverse that molecular defect without impairing mitosis [1, 7].

Sequencing identified two heterozygous candidate variants, BUB1B c.2210T>G (p.Leu737Ter) and c.3006T>G (p.Asn1002Lys). Their physical phase remains unknown, but the companion Track 1 diagnostic report quantifies how strongly the evidence favors each arrangement. Starting from equal odds, the Bayesian argument asks which configuration is compatible with an affected child: on opposite copies (trans), the child would carry one prematurely terminated allele and one missense allele, so two defective BUB1B copies, the established recessive mechanism of MVA1; on the same copy (cis), the early stop destroys that transcript and one intact copy remains, leaving the child a carrier who should not be affected. Weighing the observed MVA1 phenotype against these two predictions gives a posterior of roughly 86-95 percent for the opposite-copy configuration under the stated assumptions, conditional on BUB1B being causal for the phenotype. Parental genotyping, long-read phasing, or RNA evidence would settle the phase directly, and is the first decision point in the experimental program. Until then, I cannot declare compound heterozygosity or use these variants as proof of the diagnosis.

Why suspect RNA loss from p.Leu737Ter? The substitution creates a premature termination codon. In the analyzed BUB1B transcript (ENST00000287598), the stop lies 74

nucleotides before the exon 17/18 junction and 747 spliced nucleotides upstream of the last exon junction, as calculated in the companion variant analysis. Mammalian nonsense-mediated RNA decay (NMD) commonly recognizes transcripts whose premature stop is at least roughly 50-55 nucleotides upstream of a downstream exon-exon junction. This positional rule makes degradation plausible, but it is not universal: alternative splicing and cellular context can alter the result [3, 21]. The sequence therefore predicts susceptibility to NMD; only allele-specific RNA measurement can show whether that transcript is actually depleted in this patient.

Why suspect p.Asn1002Lys is unstable? Replacing asparagine with lysine changes residue chemistry in the C-terminal region of BUBR1. I evaluated the substitution using two different structure-based methods: FoldX, an empirical energy function, and ThermoMPNN, a learned stability predictor [4, 5]. For the project structure, both predicted destabilization: +3.61 and +1.51 kcal/mol, respectively (using the tools' reported destabilizing sign convention). The estimates differ substantially in magnitude. They do not measure BUBR1 half-life, concentration, binding, or checkpoint activity and do not establish that the proteasome clears the mutant protein. I retained both methods as a cross-check on the direction of a hypothesis, not as two votes proving pathogenicity. A third stability calculator would add less information than a patient-cell protein measurement [22].

Table 1. Each variant suggests a possible molecular defect; the rightmost column identifies the evidence required before treating that defect as real.

Finding	Reason for hypothesis	Missing evidence
p.Leu737Ter	Early stop is well upstream of a downstream exon junction; an established NMD trigger [3, 21].	Allele-specific RNA abundance and evidence of full-length versus truncated product.
p.Asn1002Lys	Two different algorithms predict destabilizing folding-energy changes [4, 5].	Mutant BUBR1 protein abundance, turnover, localization and activity.
Two variants, phase unknown	Cis and trans phase imply different combinations of normal and altered gene copies.	Parental or molecular phasing; clinical variant interpretation.

This distinction determines the next step: genetic information suggests several possible defective mechanisms, but it does not identify which mechanism a drug needs to correct. I therefore used a biochemical model to test alternative defects to create a hypothesis driven strategy for a drug search.

2. Why model the spindle checkpoint instead of searching for drugs immediately?

During mitosis, an unattached kinetochore helps generate the mitotic checkpoint complex (MCC). BUBR1 contributes to this inhibitory machinery, which restrains APC/C-CDC20 and delays securin degradation and chromosome separation. If the inhibitory machinery is weakened, checkpoint release can occur earlier; simply delaying mitosis by any means is not necessarily beneficial, because excessive arrest is itself harmful [1, 2].

I used the biochemical SAC network published by Henze and colleagues because it represents these linked reactions rather than treating BUBR1 as an isolated target. I implemented its deposited BioModels network and checked that the implementation matches those deposited reactions [2, 20]. Its job here is to answer a limited but important question: if BUBR1 abundance or a checkpoint reaction is changed, how do MCC signaling, APC/C activity and the timing of securin loss respond? It is not an existing model of aneuploidy or of drug efficacy; accordingly, exact time constants and restoration multipliers remain model-dependent.

Several published quantitative models of the spindle checkpoint exist, and I selected the Henze network for a specific fit rather than general superiority. Doncic et al. (2005) evaluated which minimal mechanisms can explain checkpoint activation, without a reaction-level network of all checkpoint proteins [29]. Ibrahim et al. (2008) modeled Cdc20 sequestration, MCC formation and APC/C inhibition with eleven species, but represented kinetochore attachment with a switching parameter rather than individual kinetochore states [27]. Ibrahim (2015)

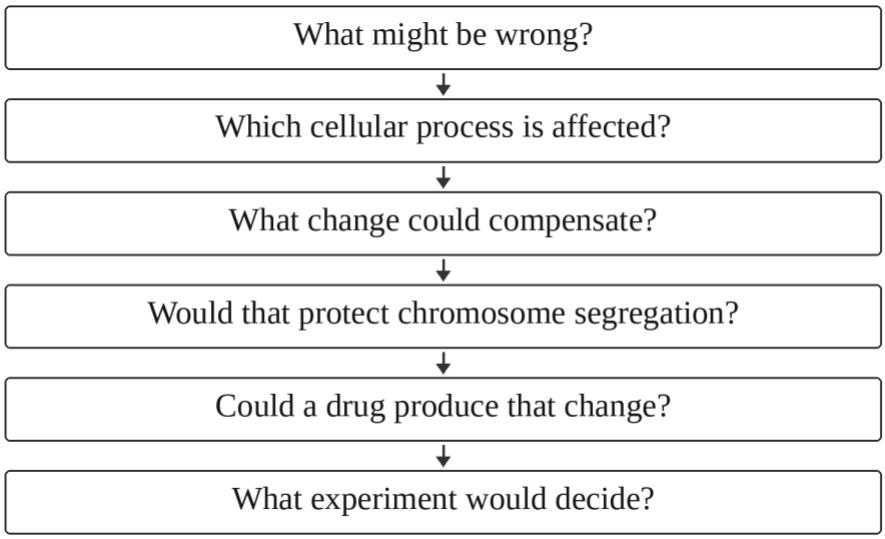
compared five inhibitory pathways and showed that tight APC/C inhibition requires both the MCC and excess BubR1, a result that supports the pool-conserving perturbations used here [28]. Chen and Liu (2014) explained why silencing waits for the final attachment with a spatial transport model in which the checkpoint signal is one distributed species rather than separate checkpoint proteins [30]. Each of these models is well suited to its own question. The Henze network is the best fit for the question in this project, how BUBR1 abundance or a specific BUBR1-dependent reaction changes MCC signaling, APC/C activity and the timing of securin loss, because BUBR1 is a first-class molecular species with explicit assembly and APC/C binding reactions, activation and silencing are driven by explicit kinetochore states, and the network is deposited as a reproducible SBML model that I independently reimplemented and verified [2, 20].

I introduced the variants as scenarios because patient-specific protein abundance and enzymatic or scaffold activity are unknown. I changed the available BUBR1-BUB3 pool parameter to represent less protein and, separately, reduced individual model reaction activities as possible consequences of the missense variant. One representative trans scenario, D1, uses a BUBR1 pool of 0.055 micromolar compared with the deposited reference of 0.220 micromolar, alongside an assumed activity impairment. More severe and abundance-only scenarios test whether the ranking depends on this guess. These values are simulations and not empirical measurements. Their purpose is to avoid building a drug recommendation on an unverified molecular interpretation.

Because p.Asn1002Lys has not been functionally characterized, the activity reductions above are three alternative hypotheses rather than one prediction, each placing a possible missense defect at a different step of the same pathway. A1, impaired MCC assembly, reduces

the rate constant k5 at which the BUBR1-BUB3 complex joins the checkpoint complex, affecting reactions 8 and 15 of the deposited network. A2, impaired BCC assembly, reduces the rate constant k7 at which BUBR1-BUB3 joins Cdc20 to form the BCC complex, affecting reaction 12. A3, impaired MCC binding to APC/C, reduces the rate constant k2 at which the assembled complex binds the APC/C, so the checkpoint machinery is produced but cannot attach to its target, affecting reactions 5 and 16. Together with the abundance and phase scenarios, these three hypotheses define the 76-scenario grid: four abundance-only runs plus 36 trans and 36 cis runs across three missense-abundance levels, four activity-reduction levels and the three hypotheses. Testing all three prevents the analysis from assuming that protein destabilization is the only possible defect of the missense variant.

Figure 1. The questions that organize the clinical inference. An uncertain answer at any point becomes an experiment, rather than being silently treated as established.



3. What could compensate for the suspected defect? A correct molecular diagnosis does not identify the best intervention. For example, increasing MCC formation, reducing CDC20 availability and increasing BUBR1 abundance could all change checkpoint behavior, but they need not have the same consequences for normal mitosis. I first measured which variables exerted the strongest local influence on model output (sensitivity or elasticity analysis). Then I simulated larger changes in those variables, because a mechanism that is sensitive to a small perturbation may be ineffective or harmful at the level a drug could cause.

Restoration of the available BUBR1 pool was the leading favorable intervention across the principal tested disease scenarios. Increasing available open MAD2 or MCC-assembly kinetics produced smaller improvements. Increasing CDC20 aggravated modeled checkpoint leakage; strong suppression of APC/C-related reactions sometimes improved restraint at the cost of excessive delay. This is why the model-informed search was narrower than a list of all possible SAC targets: I sought a change that improves signaling without creating pathological mitotic arrest. These are computational rankings under explicit assumptions, and I do not declare that all other biological strategies are unsafe [2]. However, based on the published mechanism, the sensitivity analysis, and the attachment-error extension described in Section 4, there are clearly parameters that are more effective at restoring healthy chromosome segregation levels. The logical direction then is to prioritize a drug search for candidates that would realistically improve those parameters by a meaningful amount.

The next question is not merely whether increasing BUBR1 helps, but how much change would be worth pursuing experimentally. A drug that increases the protein by 5% would be unlikely to solve a modeled deficit that requires doubling it. I therefore defined a transparent, multi-outcome checkpoint recovery score: it compares inhibitory restraint, MCC signaling and

completion of mitosis with the reference simulation. I used 0.75 as a practical partial-recovery screening benchmark and 0.90 as a near-reference benchmark. They are chosen decision thresholds, not values reported to prevent aneuploidy in human cells. Their purpose is to turn target ranking into a required-effect question, which can then be compared with pharmacological evidence. Altering the chosen score would change the exact required multiplier.

Table 2. Required increases in available BUBR1 for two chosen computational checkpoints. Each row is a hypothetical scenario, not a patient-specific therapeutic requirement.

Scenario	Starting pool	Partial model recovery (score 0.75)	Near-reference model behavior (score 0.90)
D1: abundance and activity loss	0.055 micromolar	0.110 micromolar (2.00x)	0.178 micromolar (3.23x)
D2: more severe activity loss	0.0275 micromolar	0.174 micromolar (6.32x)	Not attained by raising pool to wild-type reference
D3: abundance loss only	0.044 micromolar	0.078 micromolar (1.76x)	0.126 micromolar (2.87x)

The clinically interpretable insight is that the required simulated pool remains below the reference wild-type pool of 0.220 micromolar: the objective is restoration, not necessarily overexpression. The severe D2 scenario is equally informative for a negative reason: even a normal-sized pool does not restore near-reference output if individual BUBR1 molecules are functionally impaired. That result explains why abundance and specific activity must be measured separately before prioritizing a protein-stabilizing drug.

4. Why examine chromosome segregation if the SAC model already identifies a target?

A drug-screening model can rank biochemical interventions and still optimize the wrong clinical outcome. MVA is characterized by segregation errors and not by a laboratory MCC concentration. Henze represents attachment and checkpoint signaling but does not distinguish a correctly attached chromosome from one attached incorrectly. Merotelic attachments, in which a kinetochore is linked to microtubules from both poles, can escape normal SAC detection and contribute to chromosome mis-segregation [6]. BUBR1 also helps regulate attachment and chromosome congression through PP2A-B56, so an effect limited to MCC restoration might miss an important component of disease biology [7].

To test the limits of the checkpoint-only representation, I compared it with published human BUBR1-depletion observations. In inducible HeLa knockdown experiments, the figure-derived frequency of mis-segregation remained near 3% at 13% residual BUBR1 but rose to approximately 50% at 6%, compared with approximately 2% in the control [1]. The Henze model predicted progressively weaker signaling, but did not have an error-state output that could reproduce that cellular endpoint. This is not proof that the published checkpoint reactions are wrong: it shows why a checkpoint-only output cannot be equated with chromosome fidelity.

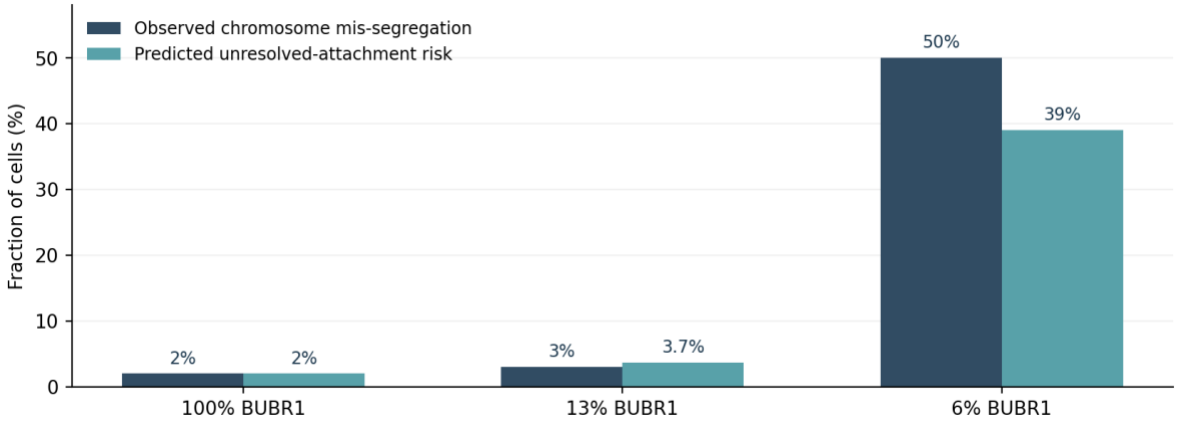
I therefore kept all the Henze checkpoint reactions and coupled them to a small project-developed attachment module. It tracks unattached, correctly attached and erroneous-but-checkpoint-silent sister-chromosome pairs, then examines whether erroneous attachments remain when the model reaches its anaphase-related event. Mechanically, the module replaces the published model's fixed attachment schedule: unattached kinetochores still generate the checkpoint signal, as in the original network, but their number now arises from capture, erroneous capture and error-correction steps rather than a fixed rate, and erroneous attachments remain invisible to the checkpoint until corrected. All other published reactions are unchanged.

This architecture is motivated by human attachment biology and detailed published spindle models, but the coarse-grained transition rates and their dependence on BUBR1 are new assumptions, not equations taken from those studies [6-8]. The purpose of the extension is twofold: to test whether missing attachment biology might explain the human depletion pattern, and to check whether an intervention that improves MCC signaling also improves a disease-related risk proxy.

5. What did the phenotype extension contribute?

The extension generated a preliminary match to the marked difference between 13% and 6% residual BUBR1. In one selected parameter setting, the modeled probability of at least one unresolved attachment error was 3.7% at 13% BUBR1 and 39% at 6%, against observed mis-segregation frequencies of about 3% and 50%. The normal-abundance observation set the model floor. Crucially, 54 settings were explored, and the displayed setting was chosen based on agreement with the observations. This is exploratory correspondence, not a prospective or independently held-out prediction. Only three of 54 settings met the prespecified within-threefold criterion for both depleted conditions.

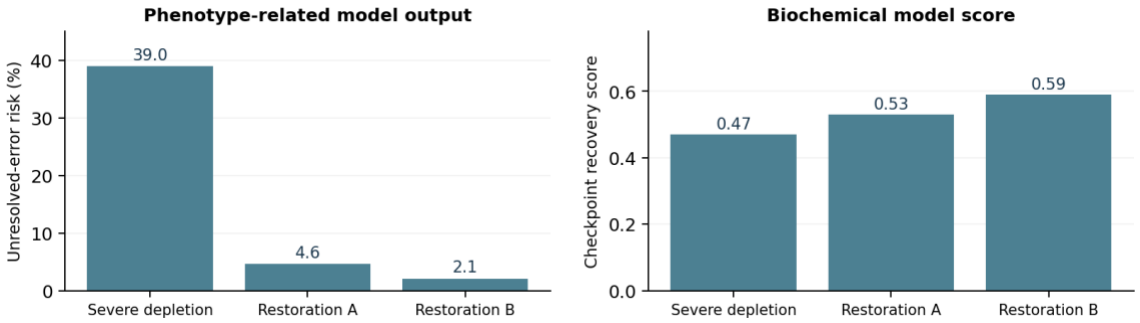
Figure 2. Human mis-segregation observations [1] and modeled unresolved-attachment risk. The quantities are related but not identical. The displayed model setting was selected after screening alternatives against the observed data.



Why test chromosome count? The modeled cell originally used 46 sister-chromosome pairs, whereas HeLa is aneuploid [9]. The probability that at least one pair has an error depends on how many pairs exist. I recalculated only that probability aggregation for 46-80 pairs; within this range both depleted observations remained within threefold of the selected model prediction and the number of qualifying parameter settings was unchanged. This does not constitute a full SAC simulation at each HeLa karyotype. It does show that the broad correspondence is not eliminated by a simple chromosome-count correction.

The key drug-discovery test was performed in a severe-depletion scenario where predicted unresolved-error risk was high enough to distinguish interventions. Raising available BUBR1 in two simulated steps reduced this model risk from 39% to 4.6% and then 2.1%; the checkpoint-recovery score rose more moderately, from 0.47 to 0.53 and 0.59. The intervention ranking was the same in the three-parameter settings compatible with the depletion observations. These are modeled molecular perturbations, not results from giving any drug. BUBR1 abundance is connected to both simulated mechanisms by design, so this agreement is an internal consistency check rather than independent evidence that BUBR1-targeting treatment prevents mis-segregation.

Figure 3. Checkpoint recovery and modeled unresolved-attachment risk respond differently to BUBR1 restoration under severe depletion. The intervention settings are not drug exposures.



This difference between outcomes matters for deciding what to test. A compound might partially improve MCC signaling yet fail to correct attachment errors; another might improve attachment fidelity without moving the combined biochemical score above the arbitrary 0.75 benchmark. I therefore retain the Henze-derived pool requirements for mechanism and potency screening, while treating chromosome alignment, lagging chromosomes and post-division aneuploidy as separate experimental endpoints that must determine whether a candidate helps the disease phenotype.

6. Could an approved medicine produce the desired molecular change?

The model identifies an intervention mechanism, but it does not automatically identify a medicine for it. I searched for approved-molecule evidence only after identifying BUBR1 restoration as the leading modeled strategy. A candidate must satisfy the entire chain: a variant really lowers functional BUBR1, a drug measurably reverses that defect, the effect occurs at biologically attainable exposure, and the restored protein improves chromosome segregation without impairing normal mitosis. Approval for an unrelated condition meets none of the variant-specific efficacy requirements.

The first indirect mechanism concerns p.Asn1002Lys: IF its destabilization causes accelerated turnover, and IF phenylbutyrate improves folding or proteostasis of that specific protein, THEN available full-length BUBR1 could increase. Sodium 4-phenylbutyrate has improved processing or function of certain unrelated mutant proteins, including CFTR and ABCA1 [10, 11]. Consequently, this establishes a reason to investigate the chemical class, not evidence of an effect on p.Asn1002Lys. I did not assume that phenylbutyrate directly blocks proteasomal BUBR1 clearance. Sodium phenylbutyrate and glycerol phenylbutyrate are both marketed for urea-cycle disorders, but the latter is a prodrug with distinct delivery and

metabolism. Their approved indications and exposures cannot be substituted for a BUBR1-specific concentration-response experiment [12, 13].

The second indirect mechanism concerns p.Leu737Ter: IF the transcript is lost through NMD, stabilizing it could increase substrate RNA; however, transcript rescue alone would not make functional full-length protein. Productive stop-codon readthrough would also have to occur, and the resulting amino acid and protein would need to function. Amlexanox has produced NMD-related and readthrough effects in other genetic systems [14, 15]. That makes it a mechanism-testing compound, not evidence that this particular stop codon is correctable. The formerly approved US amlexanox oral-paste product is discontinued [16, 17]. Under the tested literature-analogy readthrough scenarios, the predicted contribution is insufficient on its own to reach D1 checkpoint recovery; these analogies are not hard biological limits for p.Leu737Ter.

The same logic prevents an unjustified combination claim. Stabilization and readthrough could add molecules to the same BUBR1 pool, but the modeled benefit is pool arithmetic, not evidence of drug synergy, tolerability or clinical efficacy. I did not exclude all downstream or indirect therapies as inherently dangerous: the analysis identifies harmful directions within the tested SAC perturbations, and any untested drug requires its own mechanism and safety evaluation. The central pharmacological uncertainty is still whether either approved phenylbutyrate formulation increases FUNCTIONAL mutant BUBR1 at an attainable exposure, not whether other proteins have responded to high concentrations in unrelated experiments.

To turn the two candidate mechanisms into a testable comparison, I mapped which combinations of assumed drug effects would clear the model's recovery criteria in the D1 scenario (Figure 4). Because neither drug has a measured BUBR1-specific concentration-response relationship, the axes are assumed biological effects, not drug doses: the assumed

stabilization multiplier on the missense-allele BUBR1 contribution, and the assumed net functional readthrough contribution from the nonsense allele, expressed as a fraction of one normal allele. Under the modeled D1 assumptions, approximately twofold functional BUBR1 restoration alone reaches the partial-recovery benchmark, and an effective readthrough contribution of about 10% of a normal allele lowers the required stabilization to about 1.8-fold. Both legs add molecules to the same functional BUBR1 pool, so the modeled benefit is addition, not synergy; sufficient stabilization alone already meets the criterion, so the combination is not established as necessary. The map is a decision aid, not a probability that a drug works: the 1.5- to 4.8-fold stabilization range is drawn from published effects on other, unrelated proteins rather than BUBR1 measurements, and values that would push the modeled pool above the wild-type reference are mathematical extrapolations of the model, not established biological possibilities; the roughly 10% readthrough value is an illustrative effective-readthrough scenario informed by studies in other genetic contexts, not a measured ceiling for p.Leu737Ter, because ribosomal readthrough at a stop codon is not the same as functional full-length protein. On this scenario the exploratory attachment-error output is nearly flat, so drug ranking is driven by the checkpoint score; the segregation-risk comparison is reported separately in Sections 4-5.

Figure 4. Feasibility atlas for the two candidate mechanisms in the D1 scenario. Left: model checkpoint-recovery score over the assumed stabilization multiplier (vertical axis; the blue-shaded 1.5-4.8-fold box is drawn from published effects on other proteins, not BUBR1) and assumed net functional readthrough contribution (horizontal axis; the orange-shaded region up to about 0.10 reflects an illustrative scenario from readthrough studies in other genetic contexts). White dashed contours mark the 0.75 partial-recovery and 0.90 near-reference benchmarks; the red dash-dot line marks the wild-type BUBR1 pool (0.220 micromolar), beyond which the model is being extrapolated. Right: the exploratory attachment-error output, nearly flat on this scenario and reported separately in Sections 4-5. The axes are hypothetical biological effects; the measurements that decide feasibility are the BUBR1-specific assays in Section 7.

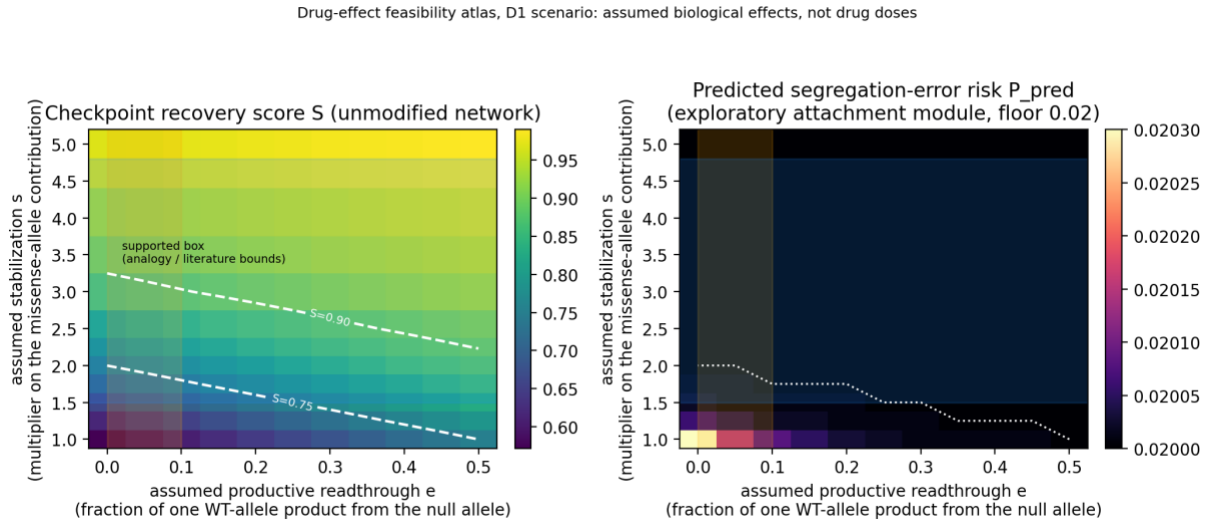


Table 3. Candidates follow from hypothesized mechanisms; none has shown target-specific treatment efficacy in this case.

Candidate	**Why investigate it?**	**What would disqualify or advance it?**
Sodium phenylbutyrate	Possible chemical-proteostasis effect; precedents in unrelated mutant proteins [10-12].	Measure mutant BUBR1 half-life, abundance and activity versus relevant exposure. Without response, deprioritize.
Glycerol phenylbutyrate	Approved prodrug source of phenylbutyrate in another disease [13].	Test its distinct exposure/metabolite profile; do not assume equivalence to sodium 4-PBA in cell assays.
Amlexanox	Prior readthrough observations in other genes [14-16].	Currently off-market in the US; require allele-specific functional readthrough, plausible exposure and safety before proceeding [17].

7. Which experiment would make the next decision possible?

Following an algorithmic precision-medicine logic, when a premise is unknown, the result should be a discriminating experiment rather than a stronger-sounding drug recommendation [23]. I ordered the experiments, so tests of genotype and molecular mechanism come before treatment-response tests. Each has a predefined interpretation that could stop work on an unpromising candidate.

Table 4. Sequential decisions and the uncertainty each one is designed to resolve. These studies are proposed.

Decision	Experiment and rationale	Decision that follows
Confirm genetic interpretation	Phase both variants and review clinical-genetic evidence; molecular mechanism differs in cis versus trans.	If not trans, rebuild the patient scenarios before any compound ranking.
Confirm NMD and protein abundance	Allele-specific RNA measurements and mitotic-phase-controlled BUBR1 protein quantification, with appropriate controls.	Retain or reject abundance-deficiency assumptions; determine whether the nonsense RNA is actually depleted.
Separate quantity from activity	Expression-matched WT versus p.Asn1002Lys complementation; checkpoint and attachment readouts at matched protein abundance.	If mutant activity is impaired despite restored abundance, stabilization alone may not correct the defect.
Test the phenotype-specific drug mechanism	Sodium 4-PBA and separately appropriate glycerol-PBA/metabolite exposures: concentration-response, protein turnover, target abundance and BUBR1 function.	Advance only if functional mutant BUBR1 increases at plausible exposure without adverse mitotic effects.
Test nonsense readthrough only if warranted	Identify allele-specific full-length protein and its function; RNA stabilization or an undifferentiated full-length immunoblot band is insufficient.	Advance or eliminate the readthrough mechanism independently of the stabilization leg.
Test disease-relevant rescue	Live-cell chromosome alignment and segregation, lagging chromosomes, and daughter-cell aneuploidy with untreated and treated controls.	Reject a candidate that improves one biochemical score but does not reduce errors or causes prolonged arrest.

A spindle-disrupting nocodazole challenge can test whether the checkpoint produces an adequate arrest in response to unattached kinetochores. It cannot, by itself, establish improved chromosome segregation while microtubules remain disrupted. To test rescue of the MVA-related outcome, I would use normal mitoses or a carefully controlled washout/recovery design with direct chromosome-error scoring. The drug hypothesis is falsified if the compound does not restore its proposed target effect, or if biochemical improvement does not yield safer chromosome separation.

8. Does this method generalize to other causes of MVA?

A method that works for one patient is not automatically a scalable precision-medicine approach. To test what could transfer, I used the same reasoning sequence in two other genetically defined forms of MVA: start with the causal gene, identify the process that gene disrupts, perturb an appropriate existing model component, and inspect both the resulting phenotype-related output and what the model cannot explain. This matters because returning the same BUBR1-restoration recommendation for different mechanisms would be misleading [18, 19]. The purpose of these small pilots was to test the workflow and expose its limits, not to nominate treatments for additional patients.

For CEP57-associated MVA2, published studies implicate centrosome maturation and spindle organization rather than a primary loss of BUBR1 [18, 19]. Because the present attachment module cannot represent centrosome geometry, I tested increased erroneous-attachment formation at normal BUBR1 as a deliberately limited proxy. With the model's baseline error-correction rate, even a 15-fold increase in the simulated erroneous-capture fraction left unresolved-error risk at the 2% reference floor. Combining increased errors with a fourfold

reduction in the correction rate raised predicted risk to approximately 4.0%-6.6%. This result identifies correction capacity as a useful experimental discriminator, but not as a demonstrated second defect in CEP57 disease. The next informative measurements would compare centrosome and spindle structure, attachment errors, and error correction in CEP57-deficient cells.

For TRIP13-associated MVA3, the biological question is different again. TRIP13 converts closed MAD2 back to open MAD2: the transition helps silence an active checkpoint and replenishes the open-MAD2 supply needed for checkpoint activation [19, 24-26]. I first reduced the silencing rates already represented in Henze. At 50% and 25% modeled activity, commitment was delayed by 8.1% and 14.1%, showing that this component responds in the expected silencing direction. But isolated silencing loss cannot represent complete TRIP13 deficiency: patient and knockout cells can fail to activate the checkpoint because the open-MAD2 pool becomes depleted [19, 24, 25]. In a separate uncalibrated probe, reducing the initial open-MAD2 pool to 50% and 25% lowered modeled peak MCC from 0.0177 to 0.0053 and 0.0021 micromolar, respectively, while checkpoint leakage increased. The opposite effects of these two probes explain why a future quantitative MVA3 model needs MAD2 recycling and availability over time. The complete-loss silencing simulation is not a prediction of the clinical disease phenotype.

Together, these pilots demonstrate that the framework can reuse computational components while returning different mechanistic questions: BUB1B restoration, CEP57-related attachment fidelity and correction, or TRIP13-dependent MAD2 availability and checkpoint silencing. Neither additional subtype has a calibrated disease model, a validated intervention threshold, or a newly established approved-drug treatment. Their main value is to identify which

next measurements or model modules would be required before a responsible subtype-specific drug search.

Table 5. Testing scalability across three genetic causes of MVA. The CEP57 and TRIP13 rows summarize exploratory computational probes, not validated disease models or drug predictions.

Subtype and mechanism	Modeled entry point	What can be concluded
MVA1 / BUB1B Possible BUBR1 deficit	BUBR1-BUB3 abundance and checkpoint activity; attachment extension	BUBR1 restoration leads among tested mechanisms in the modeled scenarios; no drug-specific rescue established.
MVA2 / CEP57 Centrosome and spindle defects	Attachment-error formation and correction proxies at normal BUBR1	Intact modeled correction buffers increased errors; true CEP57 spindle geometry is outside this module.
MVA3 / TRIP13 MAD2 remodeling	SAC silencing-rate gate and a separate initial open-MAD2-pool probe	Silencing and activation-capacity probes move outputs in different directions; full TRIP13 deficiency is not represented.

Discussion and conclusion

The inference is deliberately conditional. The variant positions and computational stability predictions justify experiments; they do not establish that the child has low functional BUBR1. The Henze network identifies which biochemical perturbations matter within a published reaction system, and its chosen checkpoint score is not a clinically validated aneuploidy threshold. The attachment extension adds a disease-related risk proxy and exploratory agreement with human depletion observations, but its functional laws were not independently identified, and it was selected against the observed outcomes.

The actionable result is a ranked causal hypothesis. IF the variants are in trans, IF p.Asn1002Lys chiefly lowers available but otherwise usable BUBR1, and IF a clinically

attainable exposure can increase that functional pool, THEN BUBR1 restoration is predicted to improve modeled checkpoint signaling and may reduce unresolved chromosome-attachment risk. Each IF represents a distinct measurable premise. A negative result at any point is useful: it redirects the search toward activity defects, attachment-specific mechanisms or alternative molecular targets rather than promoting an ineffective drug.

Therefore, I conclude that BUBR1 restoration is the leading tested intervention mechanism for the modeled BUB1B scenarios, while phenylbutyrates and amlexanox remain conditional laboratory hypotheses. The contribution is a reproducible route from genomic uncertainty to a mechanistic intervention and the experiments required to learn whether that intervention is biologically and pharmacologically achievable. The CEP57 and TRIP13 pilots separately demonstrate that this reasoning workflow can be reused for other MVA mechanisms, while also identifying missing disease biology that prevents justified subtype-specific drug predictions.

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Technical supplement and reproducibility

The companion repository contains the exact Henze SBML and software-verification tests, deposited-versus-published version analysis, model scores and scenario definitions, computational variant-stability results, extended-attachment equations, 54-setting screen, chromosome-count sensitivity, intervention matrices, pharmacological evidence dossier, and proposed patient-cell decision gates. The separate docs/MVA2_MVA3_Scalability_Pilot.md and results/Scalability_MVA2_CEP57.csv / results/Scalability_MVA3_TRIP13.csv document the exploratory subtype pilots, the TRIP13 activation-versus-silencing correction, the incomplete CEP57 spindle representation, and null or artifact-prone results.